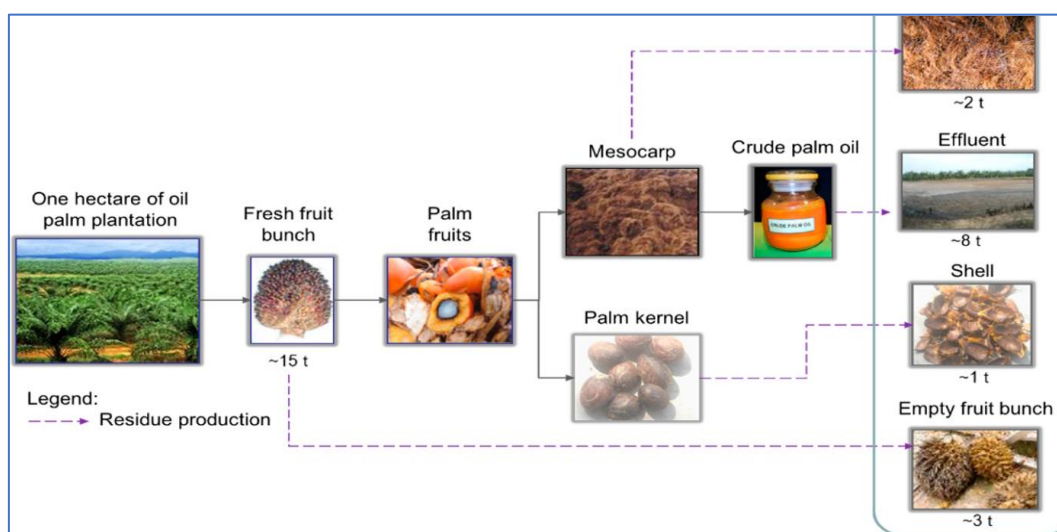


pharmaceutical products (Casson, Muliastira, and Obidzinski, 2014). The development of biofuels from biomass has raised interest in expanding the palm oil plantation area. This is because palm oil is the main raw material for biodiesel in Indonesia.

CPO is the primary product derived from the red fruit of the oil palm, while palm kernel oil, derived from the fruit's nut, is considered a secondary product. Oil palm biomass includes EFBs, palm mesocarps fibres (PMFs), PKS, oil palm fronds, oil palm trunks, as well as palm oil mill effluent (POME). Oil palm fronds account for 70% of the total oil palm biomass produced, while EFB accounts for 10% and oil palm trunks account for only about 5% of the total biomass produced.

According to Harahap et al. (2019), Indonesia housed 11 million hectares (Mha) of oil palm plantations and produced 31 million tonnes (Mt) of CPO in 2015. Oil extraction from palm fruits occurs in palm oil mills. One tonne (t) of CPO production results in nearly 5 t of solid biomass waste, including EFBs, PKSs, PMFs, and POME; see Figure 3.3. This implies that, in 2015, Indonesia produced around 155 Mt of palm biomass residue.

Figure 3.3. Biomass Use in Oil Palm Industry



Source: Harahap et al. (2019).

Regarding the potential for biodiesel, the previous Table 2.10 projected the demand of FAME for both B30 and B40 mandates using the volume of diesel fuel needed for the road transport sector. As shown, the FAME demand will reach 19.1 million kL in 2040 for the B30 mandate and 25.4 million kL for the B40 mandate. The current FAME production capacity is 12.85 million kL, indicating a shortage of supply to meet the 2040 demand for both the B30 and B40 mandates.

Increasing the capacity for FAME production implies that the demand for domestic CPO will continue to increase. The estimated CPO required to produce FAME in 2040 is also calculated above (Table 2.11). The estimated CPO consumption for B30 and B40 mandate in 2040 will be 17.5 and 23.4 million tonnes, respectively. This was calculated based on